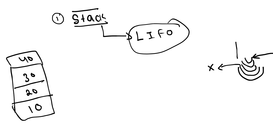
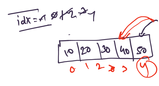
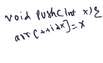
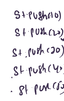
Lec-23

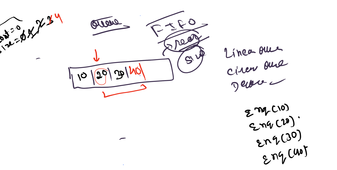
29/03/26 4:41 PM



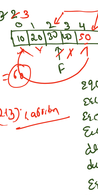


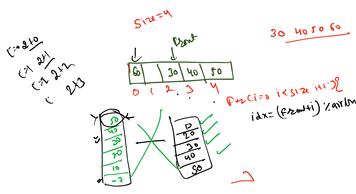


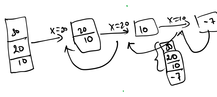




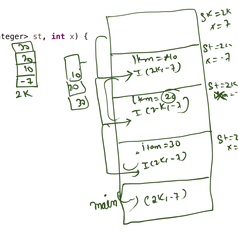

```
// O(1)
    public void Enqueue(int x) throws Exception {
        if (isfull()) {
            throw new Exception("Bklol hai tu Queue full hogya hai");
        }
        arr[front+size] = x;
        size++;
    }

    // O(1)
    public int Dequeue() throws Exception {
        if (isEmpty()) {
            throw new Exception("Bklol hai tu Queue Empty  hai");
        }
        int x = arr[front];
        front = front + 1;
        size--;
        return x;
    }

    // O(1)
    public int getFront() throws Exception {
        if (isEmpty()) {
            throw new Exception("Bklol hai tu Queue Empty  hai");
        }
        int x = arr[front];
        return x;

    }
```

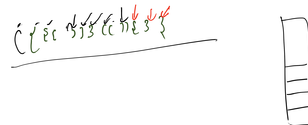







```
public static void Insert(Stack<Integer> st, int x) {
    if (st.isEmpty()) {
        st.push(x);
        return;
    }
    int item = st.pop();
    Insert(st, x);
    st.push(item);
}
```





2375. Construct Smallest Number From DI String

Solved

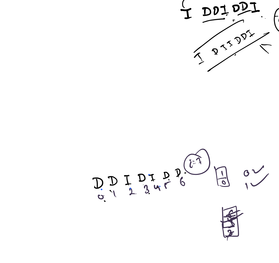
Medium Topics Companies Hint

You are given a **0-indexed** string `pattern` of length `n` consisting of the characters `'I'` meaning **increasing** and `'D'` meaning **decreasing**.

A **0-indexed** string `num` of length `n + 1` is created using the following conditions:

- `num` consists of the digits `'1'` to `'9'`, where each digit is used **at most** once.
- If `pattern[i] == 'I'`, then `num[i] < num[i + 1]`.
- If `pattern[i] == 'D'`, then `num[i] > num[i + 1]`.

Return the *lexicographically* **smallest** *possible string* `num` *that meets the conditions.*

Example 1:

```
Input: pattern = "IIDIDDD"
Output: "123549876"
Explanation:
At indices 0, 1, 2, and 4 we must have that num[i] < num[i+1].
At indices 3, 5, 6, and 7 we must have that num[i] > num[i+1].
Some possible values of num are "245639871", "135749862", and "123849765".
It can be proven that "123549876" is the smallest possible num that meets the
conditions.
Note that "123414321" is not possible because the digit '1' is used more than
once.
```